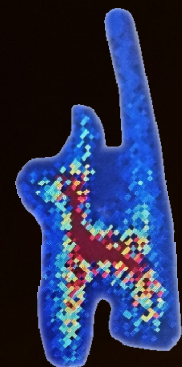
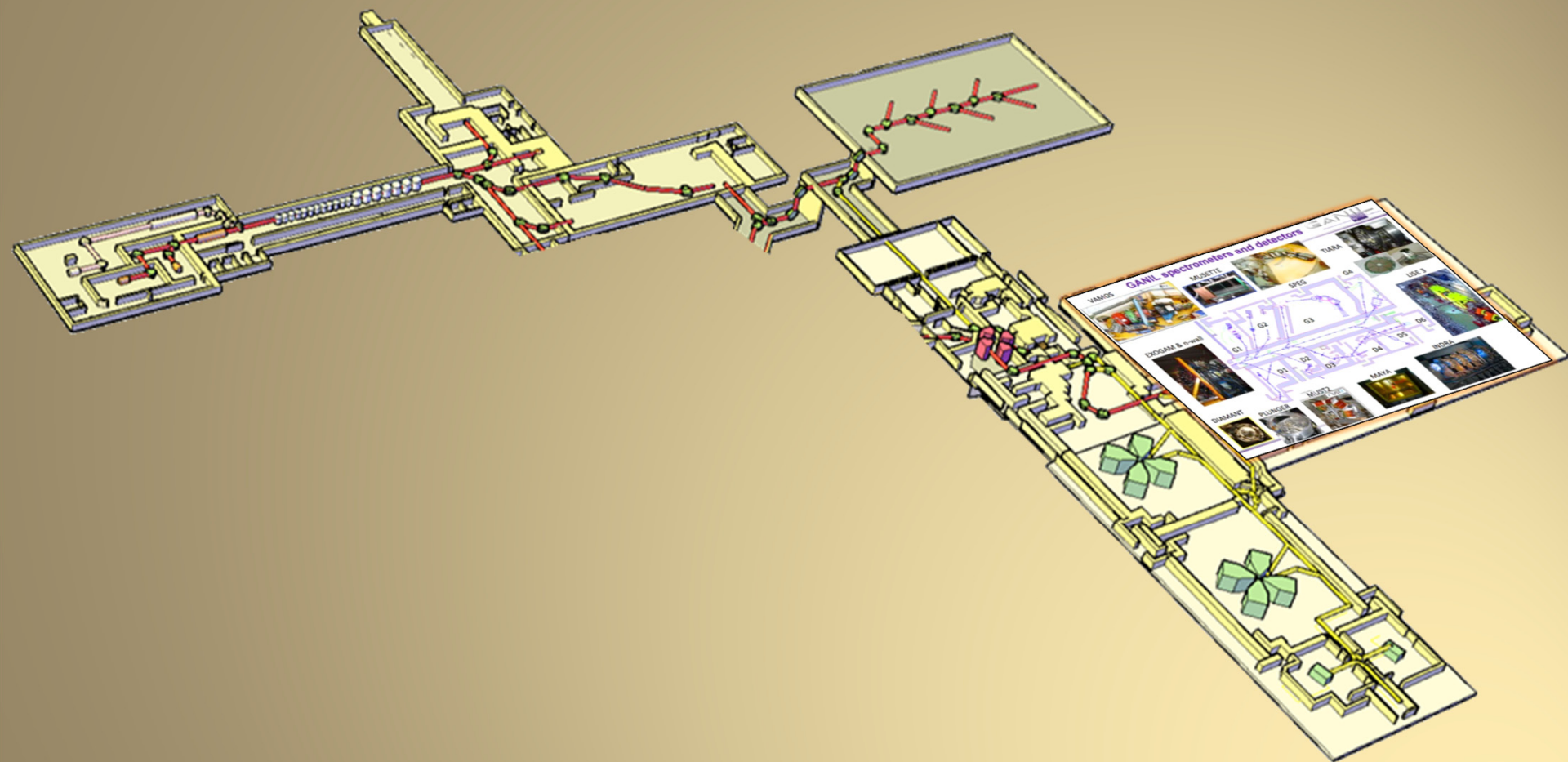




STATUS OF SPIRAL2 AND RFQ BEAM COMMISSIONING

R. Ferdinand – SPIRAL2
For the SPIRAL2 team



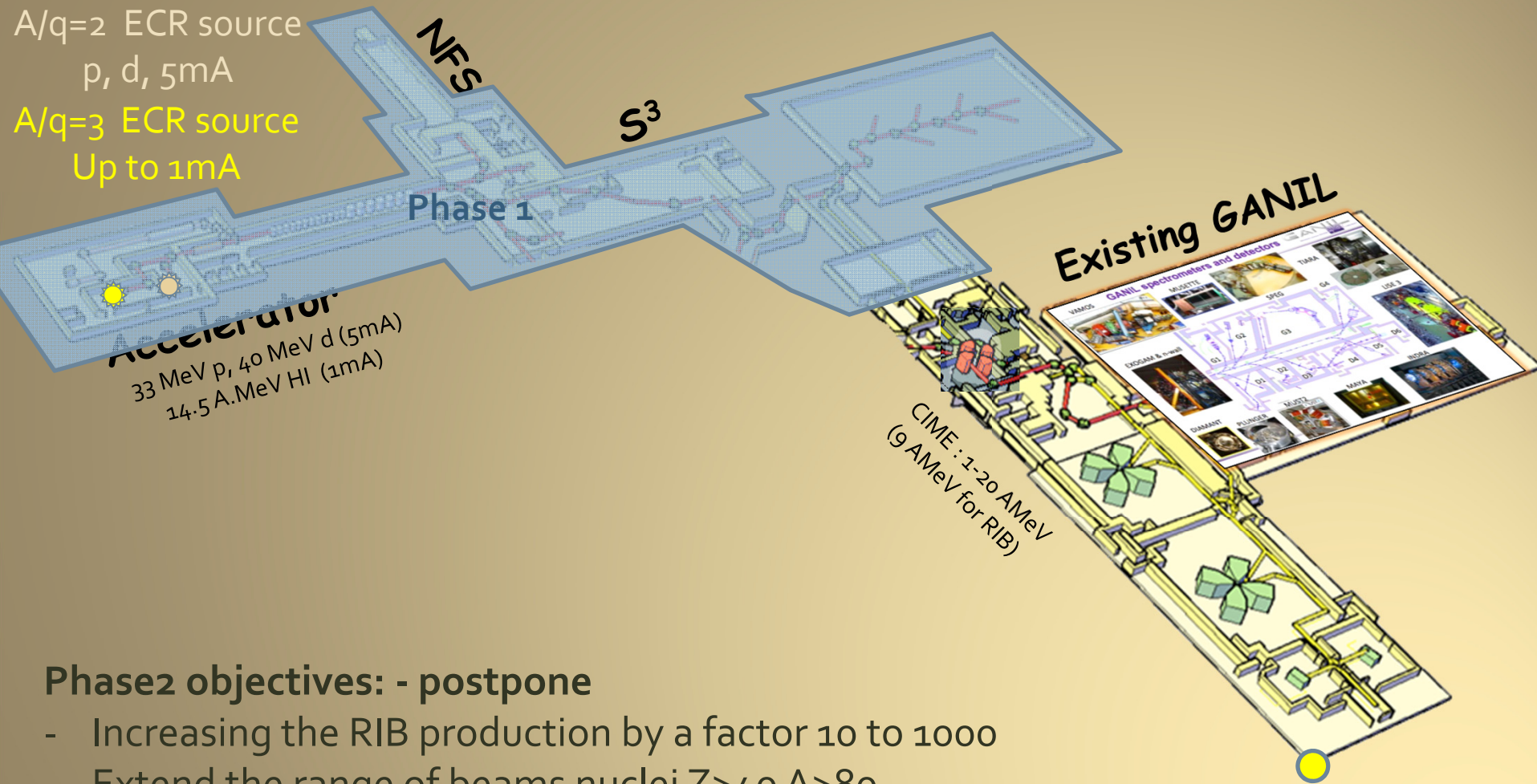


Phase1 objective:

Increasing the stable beam power by a factor 10 to 100

NFS

- Production of an intense neutron flux
- Material irradiation, cross section measurements (for ADS, gen.IV, fusion etc...)



Phase2 objectives: - postpone

- Increasing the RIB production by a factor 10 to 1000
- Extend the range of beams nuclei $Z > 40$ $A > 80$

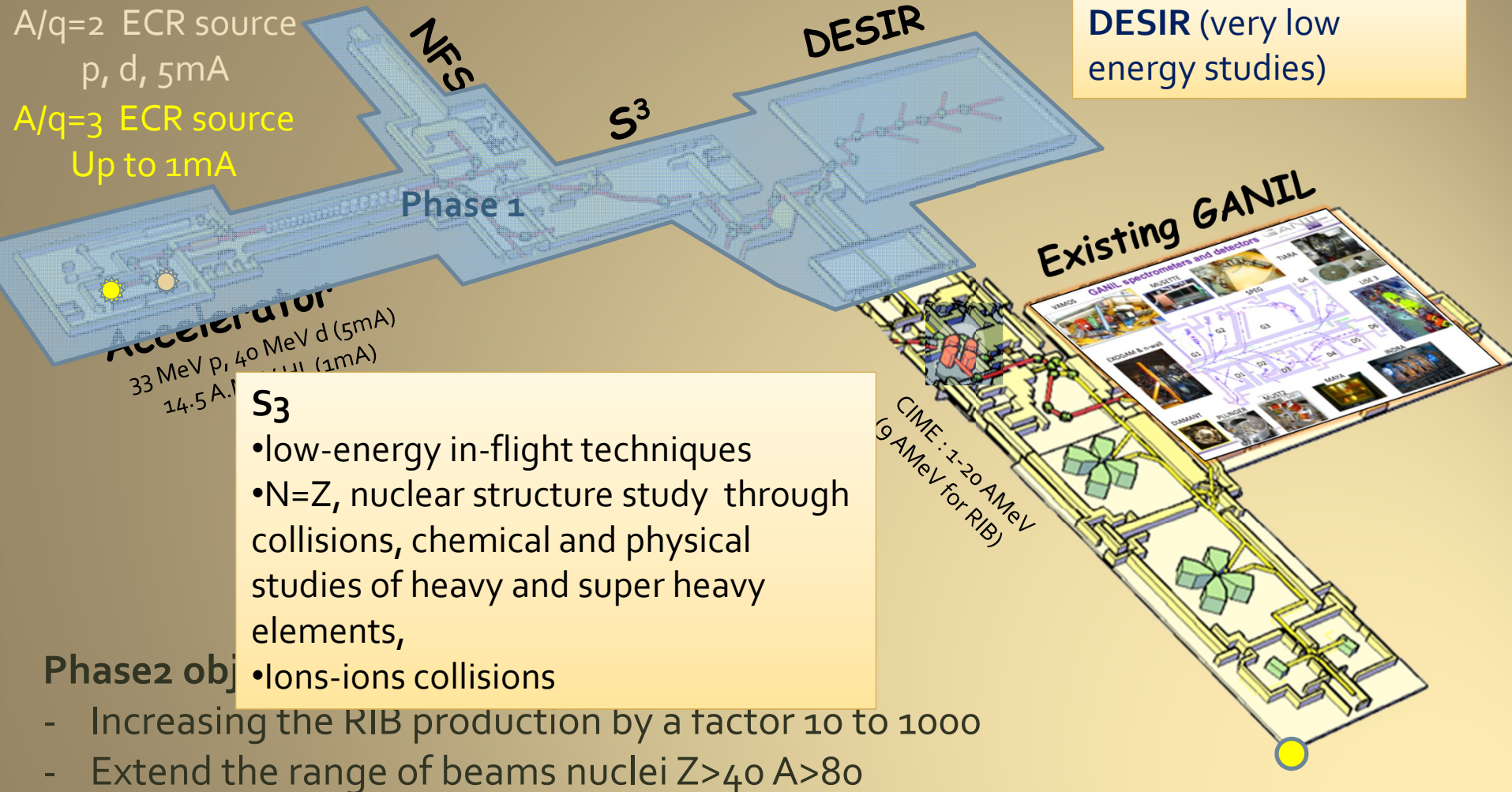
Phase1 objective:

Increasing the stable beam power by a factor 10 to 100

NFS

- Production of an intense neutron flux
- Material irradiation, cross section measurements (for ADS, gen.IV, fusion etc...)

DESIR (very low energy studies)



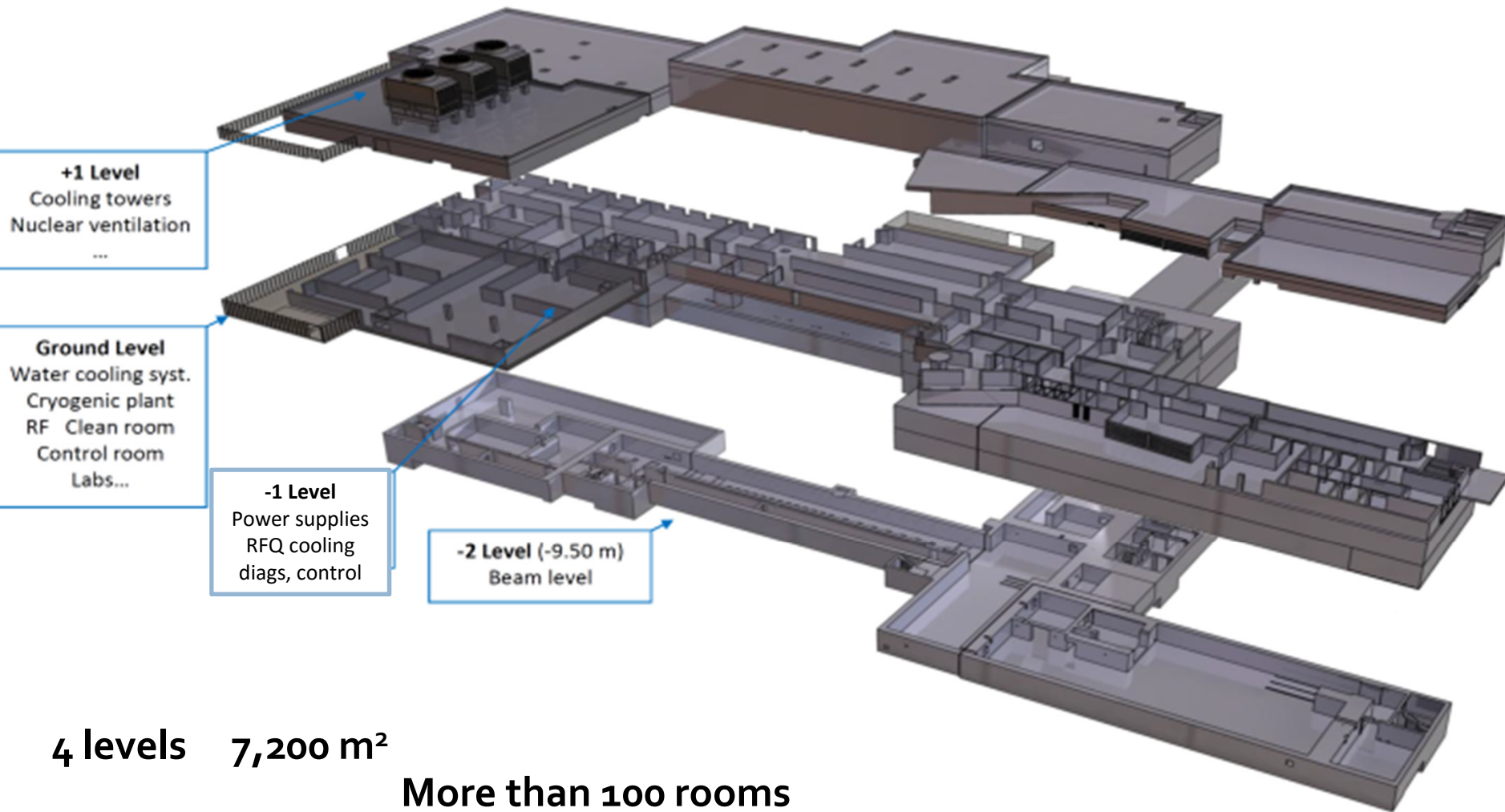
S³

- low-energy in-flight techniques
- N=Z, nuclear structure study through collisions, chemical and physical studies of heavy and super heavy elements,
- Ions-ions collisions

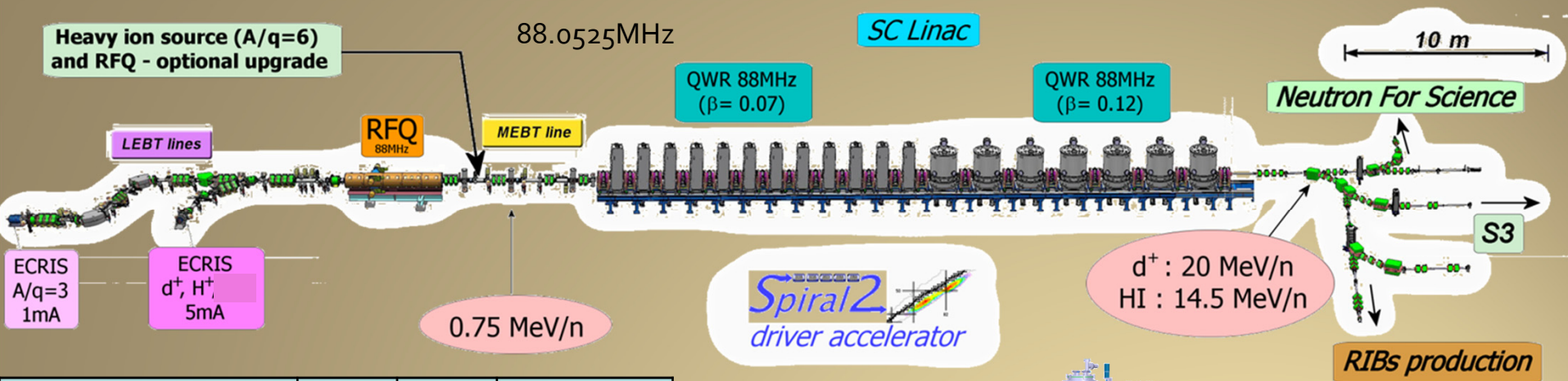
Phase2 objective:

- Increasing the RIB production by a factor 10 to 1000
- Extend the range of beams nuclei $Z > 40$ $A > 80$

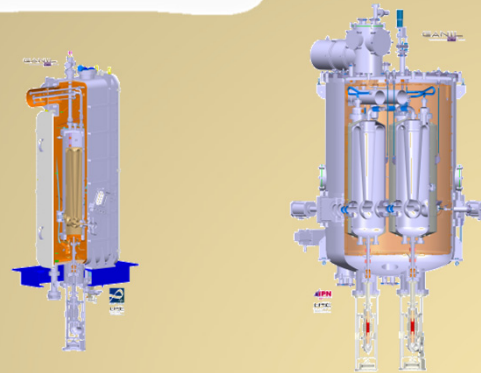
SPIRAL2 Facility



Accelerator Configuration



Particles	H^+	D^+	Ions	
Q/A	1	1/2	1/3	1/6
I (mA) max.	5	5	1	1
W_0 max. (MeV/A)	33	20	15	8.5
CW max. beam power (KW)	165	200	44	51



Total length: 65 m (without HE lines)

12 QWR beta 0.07 (12 cryomodules)
 14 (+2) QWR beta 0.12 (7+1 cryomodules)
 1.1 kW Helium Liquifier (4.5 K)
 Room Temperature Quadrupoles
 Solid State RF amplifiers (10 & 20 KW)
 6.5 MV/m max $E_{acc} = V_{acc}/(\beta_{opt}L)$ with $V_{acc} = \int E_z(z) e^{i\omega z/c} dz$.

SPIRAL2 building



Building and infrastructures Work Packages...

DONE



Excavation

Civil Engineering

Nuclear Ventilation

Conventional Ventilation

Water cooling system

Electricity

Handling equipments
and Special equipments

THPLR024 P. Anger, et al
...process integration...

09/28/2016

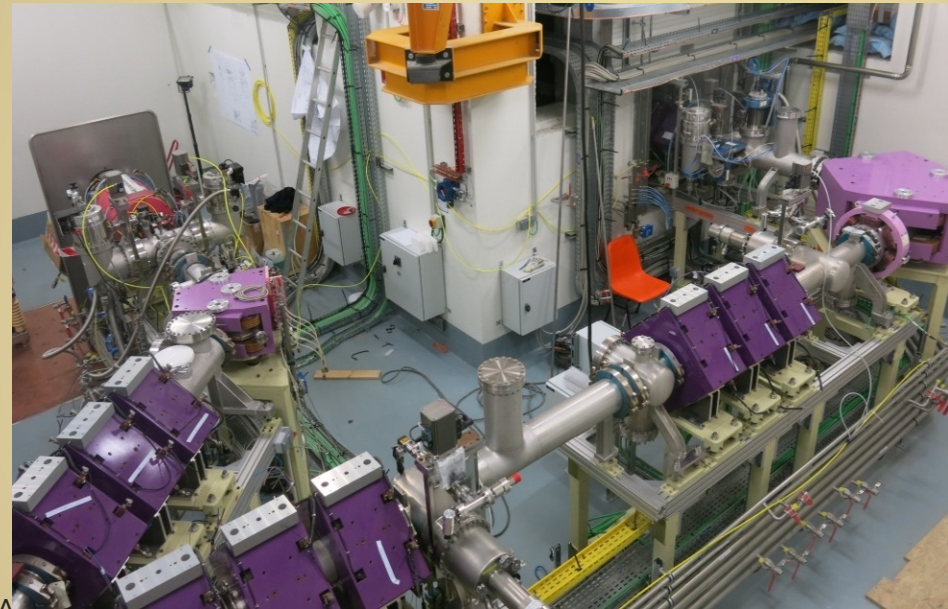
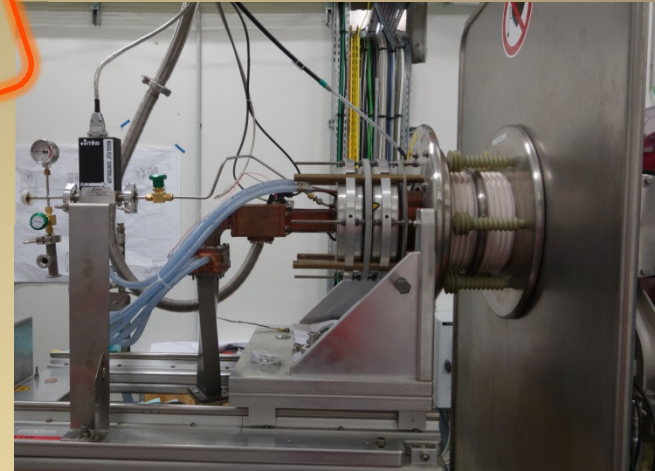
Robin FERDINAND - LINAC2016

7



Sources and LEBT

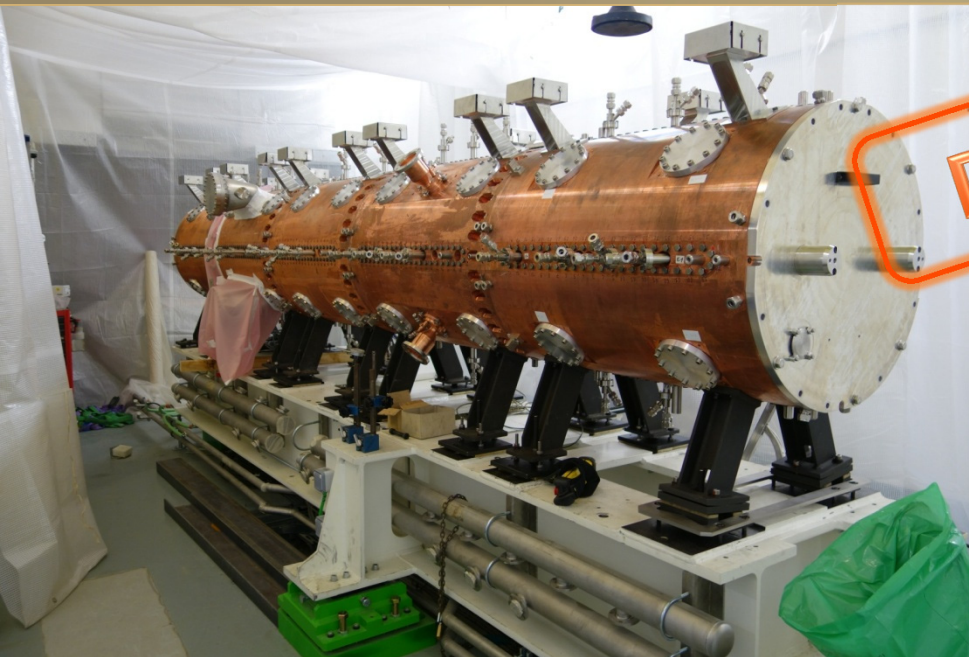
DONE



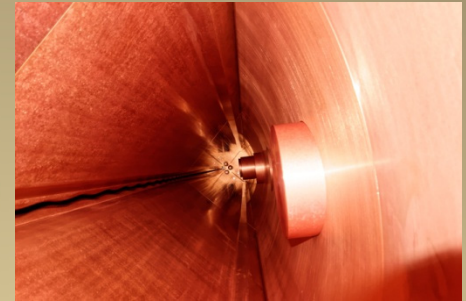
First beam ($230 \mu\text{A}$ Argon 9^+) July 10, 2015

First beam (2mA H^+) Dec. 19th, 2014

SPIRAL2 4-vanes RFO

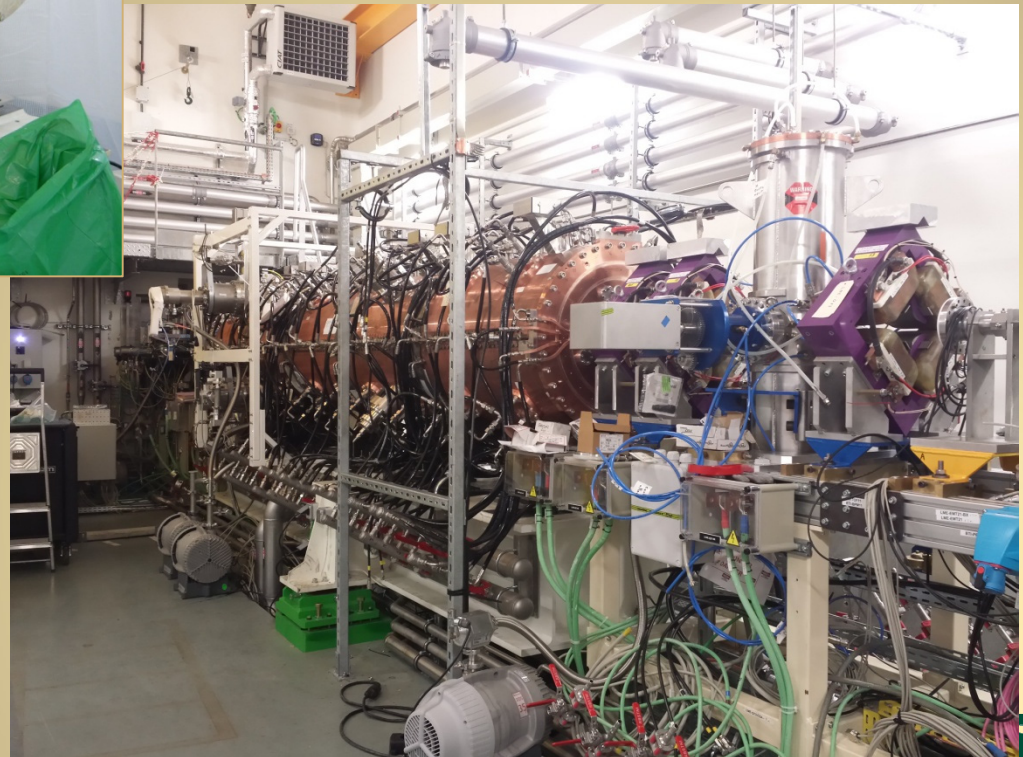


DONE



Assembly from Sept 2014 to
March 2015

First beam (H^+) Dec 3rd, 2015



SPIRAL 2 cryogenic system

Cryogenic system (1,300 W equivalent 4.5 K) installed, first liquid helium in July 2015



DONE



SPIRAL₂ LiNAC status

● Today :

■ 12 /12 CMA

■ 6 /7 CMB aligned and connected to their valve boxes

Almost DONE



All valves boxes needed reparations
(1 year delay)
Still 1 cryomodule with cryogenic leaks

First cool down July 13, 2016

SPIRAL2 linac warm sections

18 / 20 hot-sections installed



SPIRAL₂ RF amplifiers

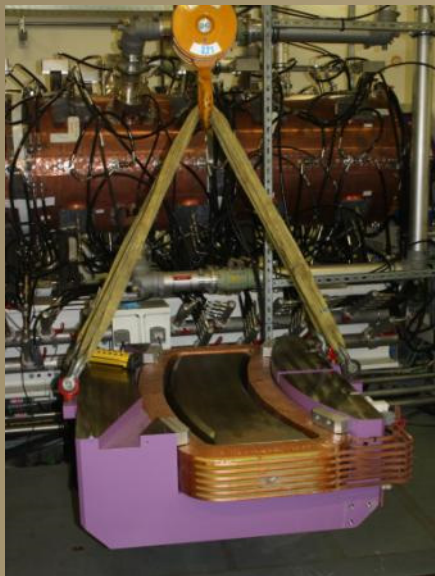


DONE

One solid state amplifier
/ cavity
Up to 20 kW

Amplifiers,
feeders, circulators,
LLRF and interlock PLC
have been tested
independently, installed
in the building and
interconnected

SPIRAL₂ HEBT line installation



Safety roles
Switch
power measurement
Beam dump

Commissioning Strategy



SPIRAL2 four commissioning phases

- qualification of the ion sources and LEBT in the laboratories in charge of the development
 - 2009-2012
- qualification of the injector on a test bench
 - Reproduce the results from the pre-commissioning
 - Validate the RFQ performances : transmission, beam energy, output emittances in the three planes, and bunch extension,
 - Provide a development platform for various diagnostics
 - Measure the beam characteristics at the RFQ exit, to serve as input parameters for the next stage.
- SC linac beam commissioning up to the main beam dump
- “day-1” experiments to NFS and S3 experimental halls, including commissioning.



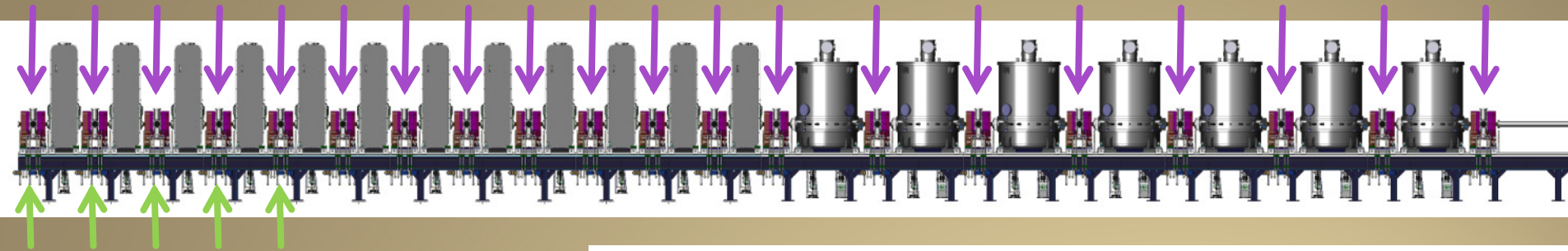
Strategy for a multi beam commissioning

● increasing stress for the RFQ cavity and later for the linac

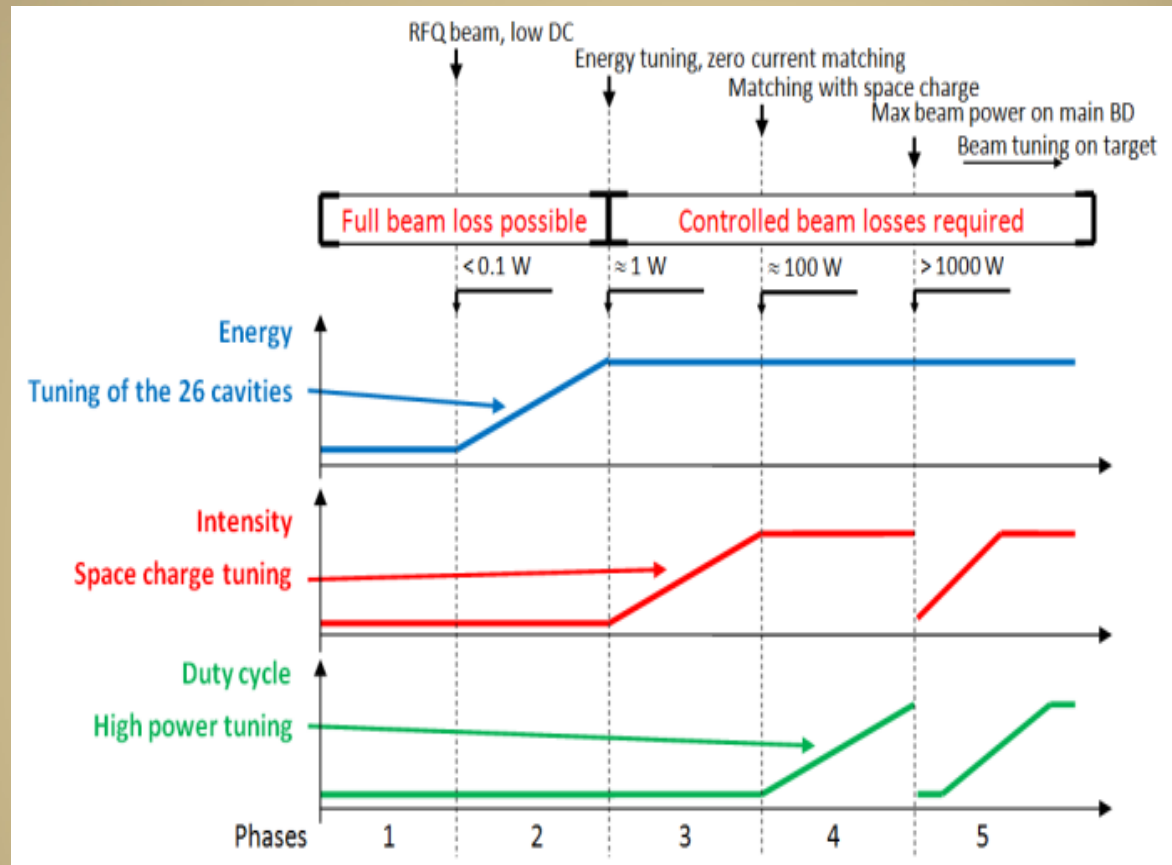
- ❶ : H^+ beam, Light ion source and LEBT, 50kV on RFQ vanes, up to 5mA
- ❷ : $^4He^{2+}$ beam, heavy ion source and LEBT, 80kV, up to 2mA, mimic the D^+
- ❸ : ultimate injector performances, $A/Q=3$, 114kV, up to 1mA $\rightarrow ^{18}O^{6+}$
- ❹ : 20 μA Ni ion beam to facilitate the future tuning of the accelerator
- ❺ : D^+ beam, up to 5mA
 - ▶ requires the final licence from the Safety Authority Offices
 - ▶ Activation



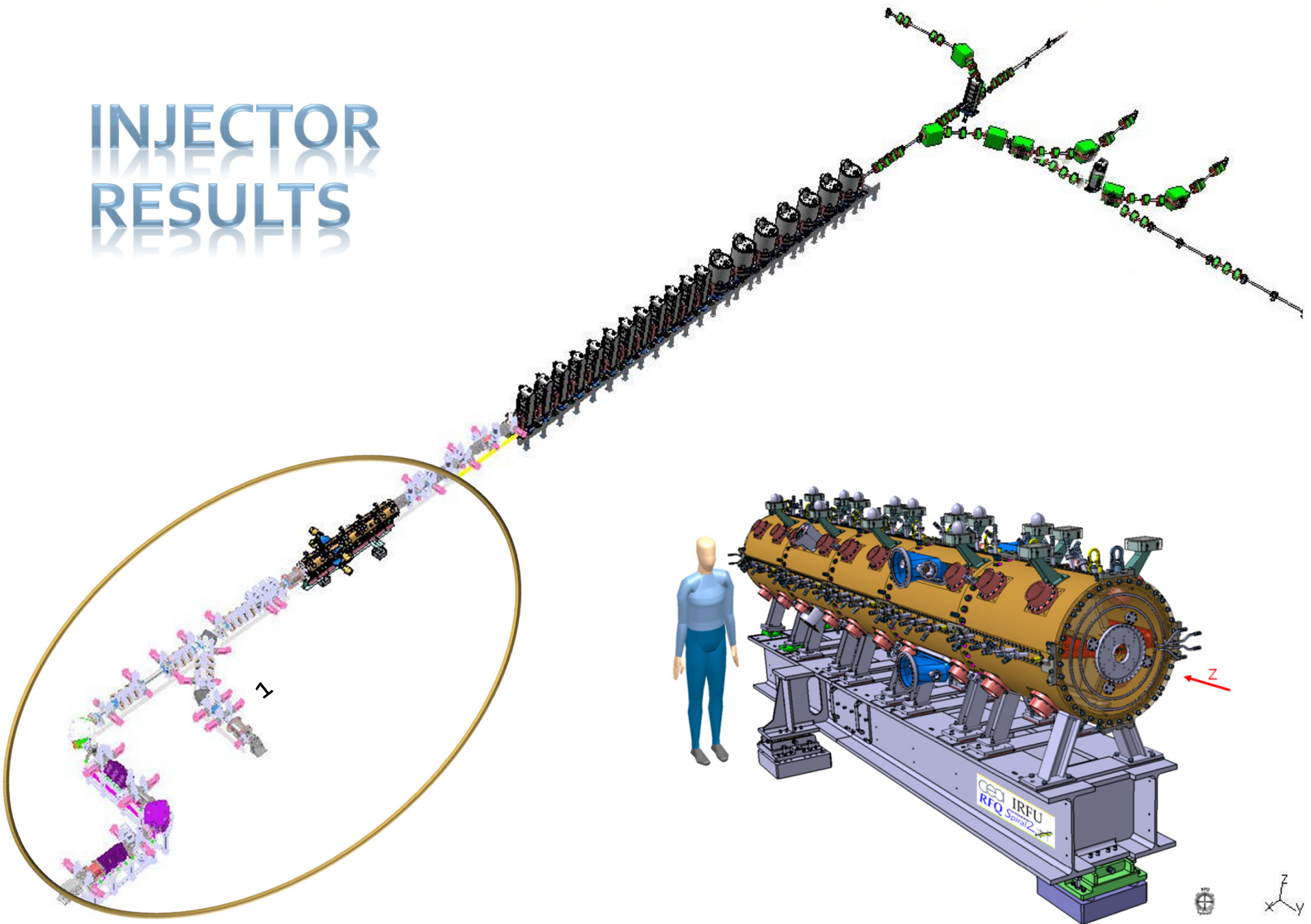
LINAC commissioning plans



- Adjust quadrupoles,
 - BPMs, BEMs
- Match the MEBT beam to the linac channel,
 - BPMs
- Tune the first linac cavity (amplitude & phase)
- Re-adjust all linac and HEBT quadrupoles,

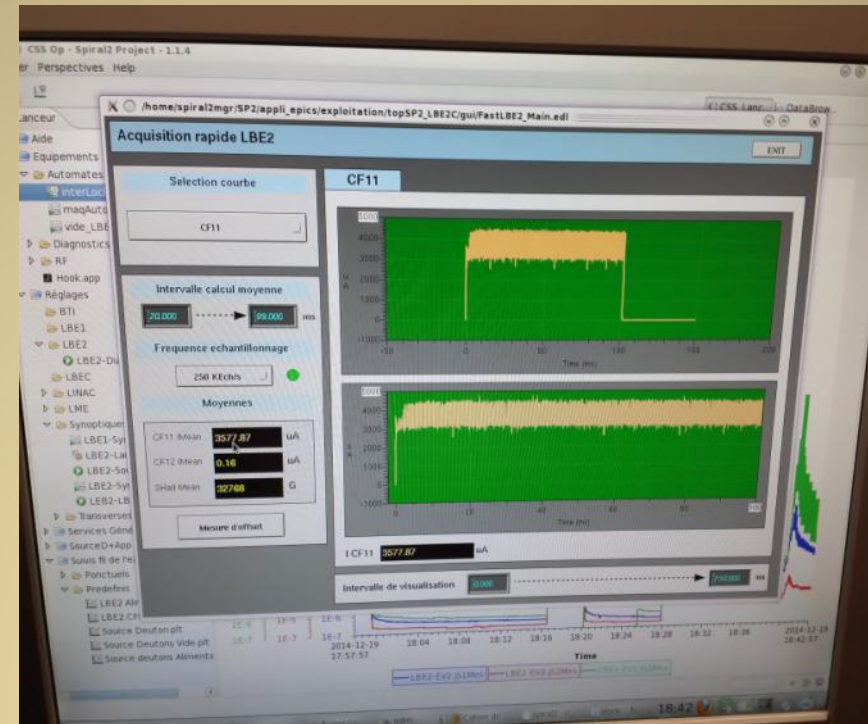
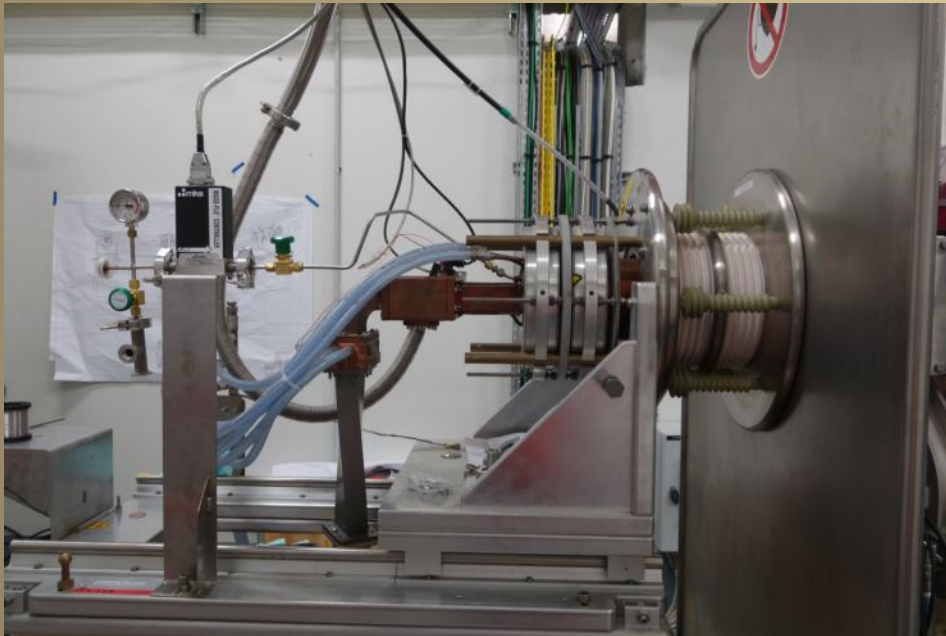


INJECTOR RESULTS



Proton/deuteron source commissioning

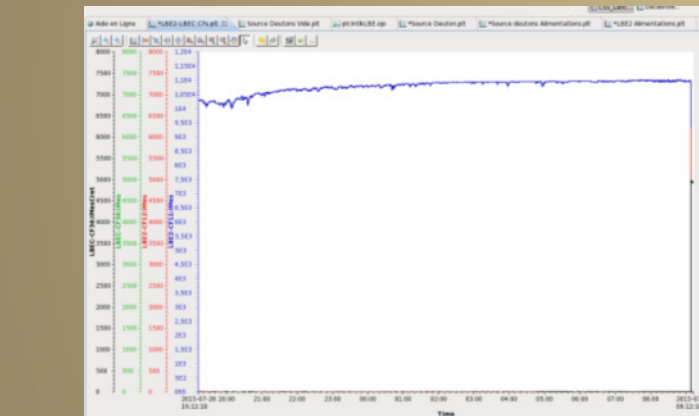
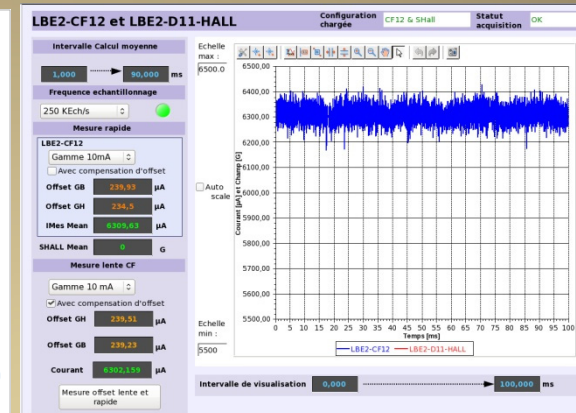
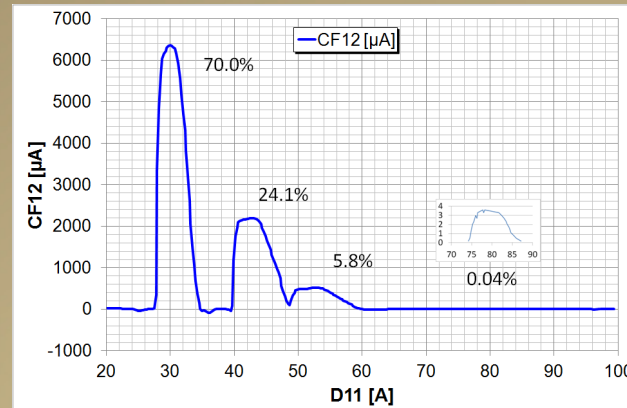
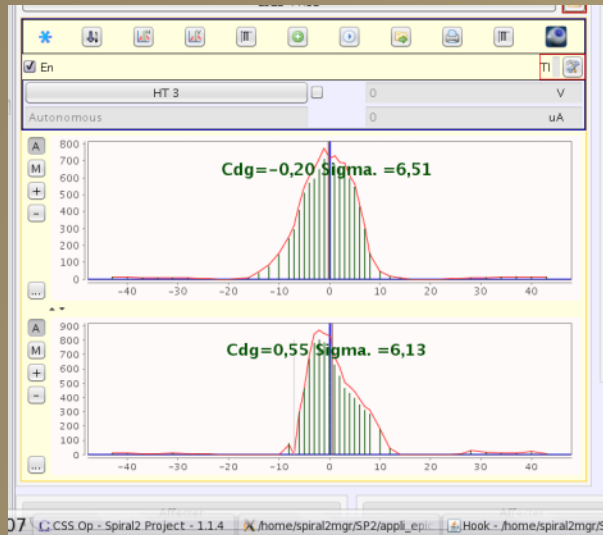
First beam at GANIL
December 19, 2014



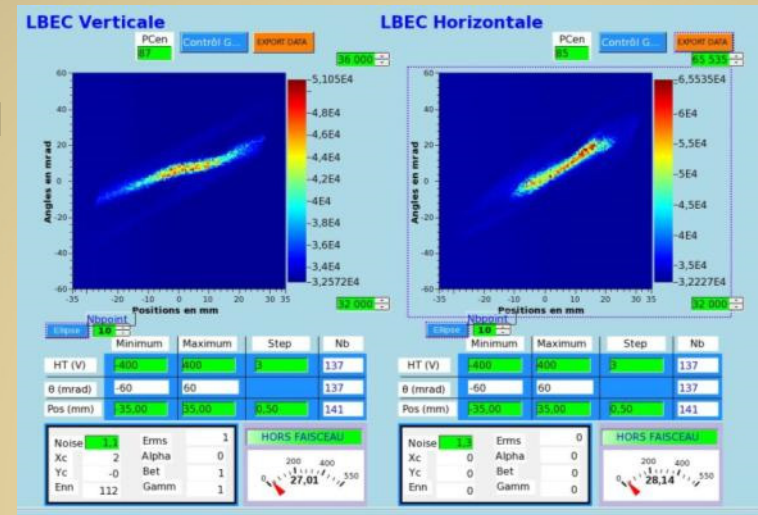
Protons

Issue = noise (emittance growth)

SPIRAL2 H/D and HI source commissioning



0.2 π mm mrd
rms norm.



Long period stability (6 mA CW)

Emittance measurements f(l) July + August + Sept

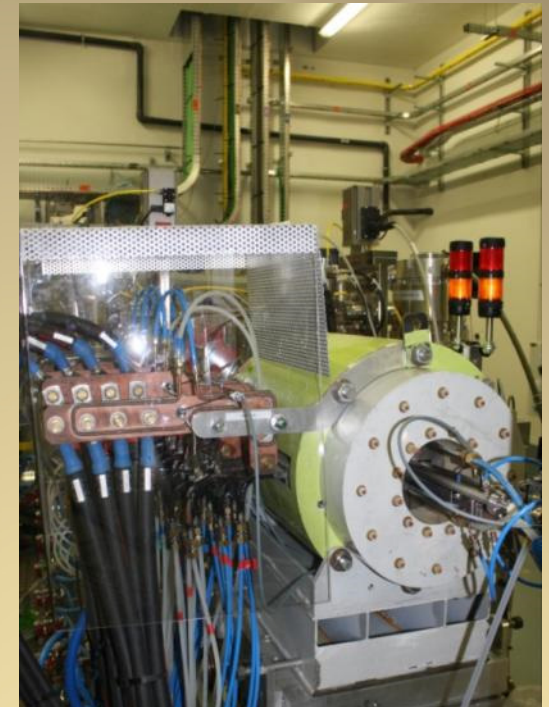
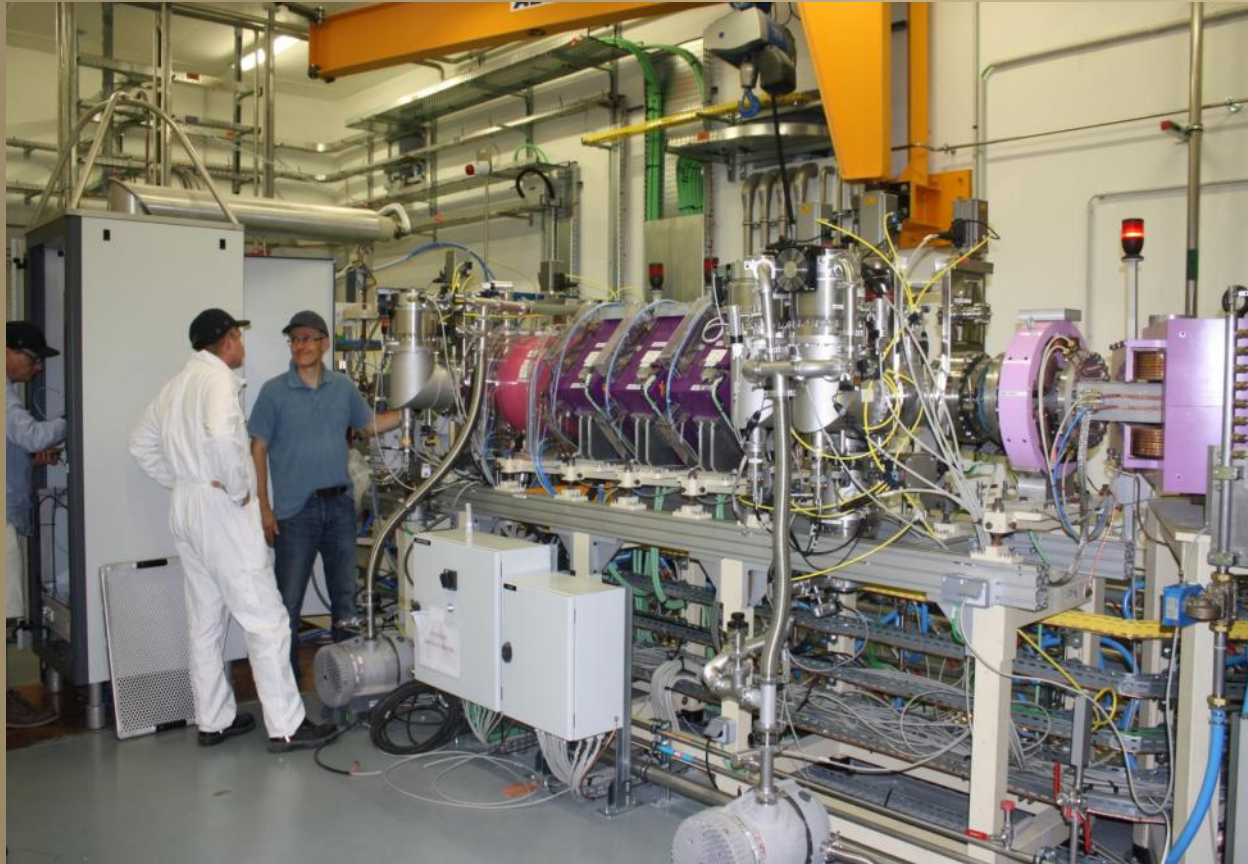
Ability to extract 11 mA CW from the p source => 6 mA proton beam

Beam intensity and emittance control using 6 H and 6 V slit systems



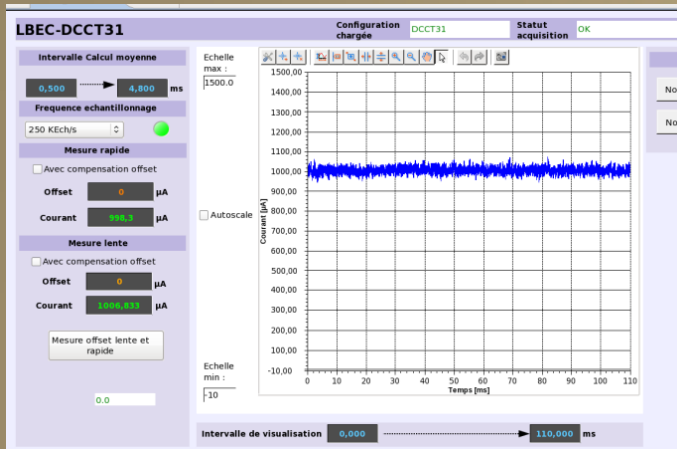
Heavy-Ion A/Q = 3 source commissioning

First beam ($230\ \mu\text{A}$ Argon 9+) July 10, 2015

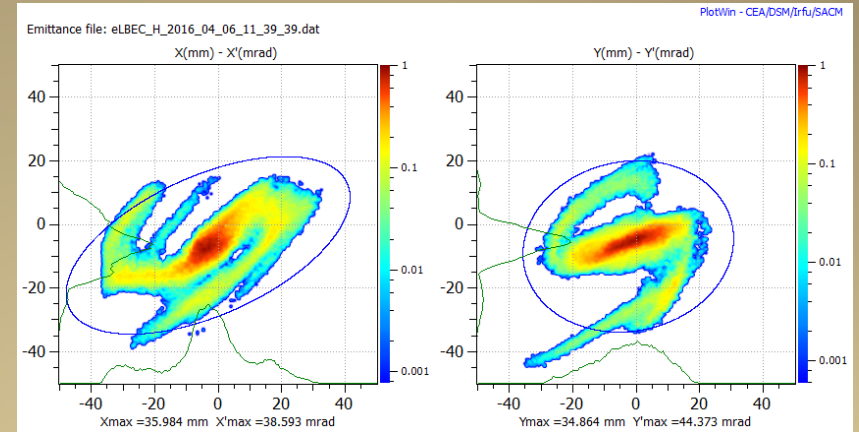


Test
 ${}^4\text{He}^{2+}$ A/Q = 2

Some results at the end of the LEBT

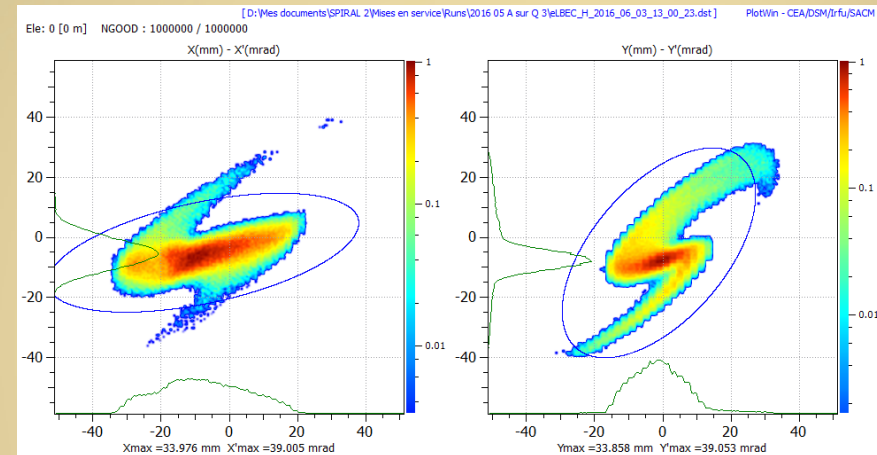


$^4\text{He}^{2+}$, 1 mA, CW



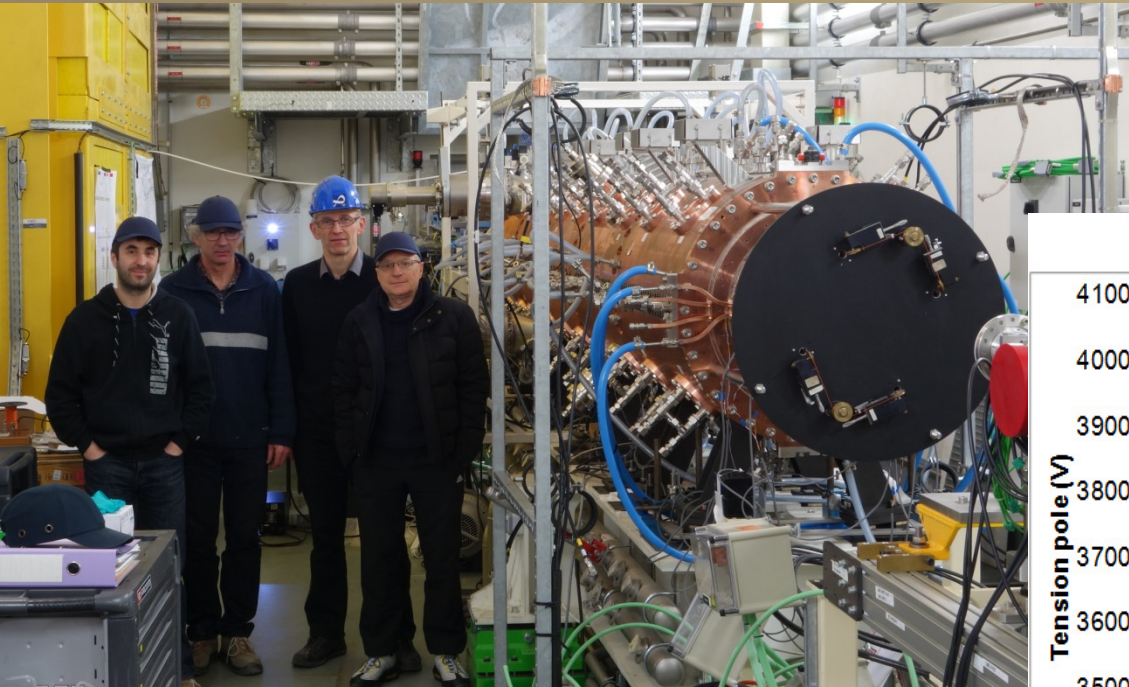
1.35 mA He, open slits,
before hexapole tuning optimization

Particle	Beam current (mA)	Emit X (π .mm.mrad)	Emit Y (π .mm.mrad)
H^+	5.2	0.18	0.2
$^4\text{He}^{2+}$	1.35	0.54	0.43
$^{18}\text{O}^{6+}$	0.75	0.36	0.42

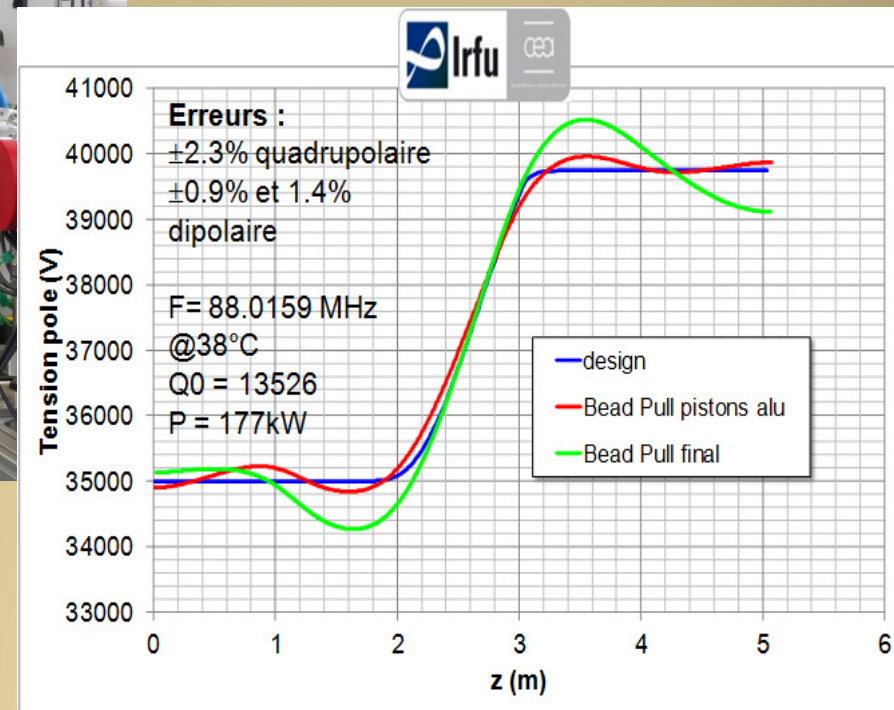


750 μA $^{18}\text{O}^{6+}$, $\epsilon=0.36/0.42 \pi$.mm.mrad

RFO Low Level tuning



IPAC16 – MOOCA02,
O. Piquet talk : RFO dev at CEA-IRFU



Voltage law (bead-pull) measurements + adjustments (40 plungers) OK March 2015
Max error = 2.3 % on the quadrupolar mode (+3 % longitudinal emittance increase)

Expected transmission accelerated particle = 99.7 %

RFQ high Power commissioning

 **4 x 6**

prea

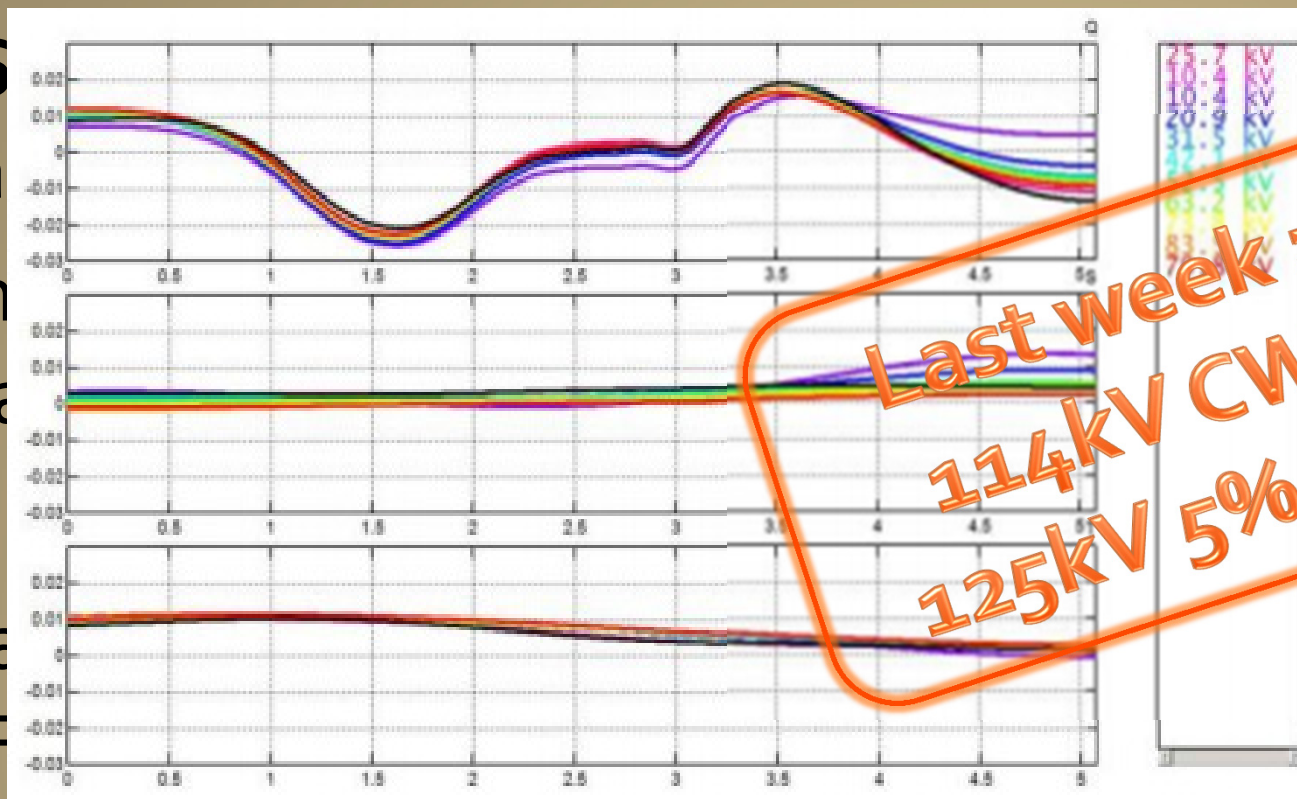
4 am

$$= = >$$


in RF

● relative errors are less than $\pm 0.2\%$ as the vane voltage is varied from 20 kV to 80 kV

Stable long term behavior

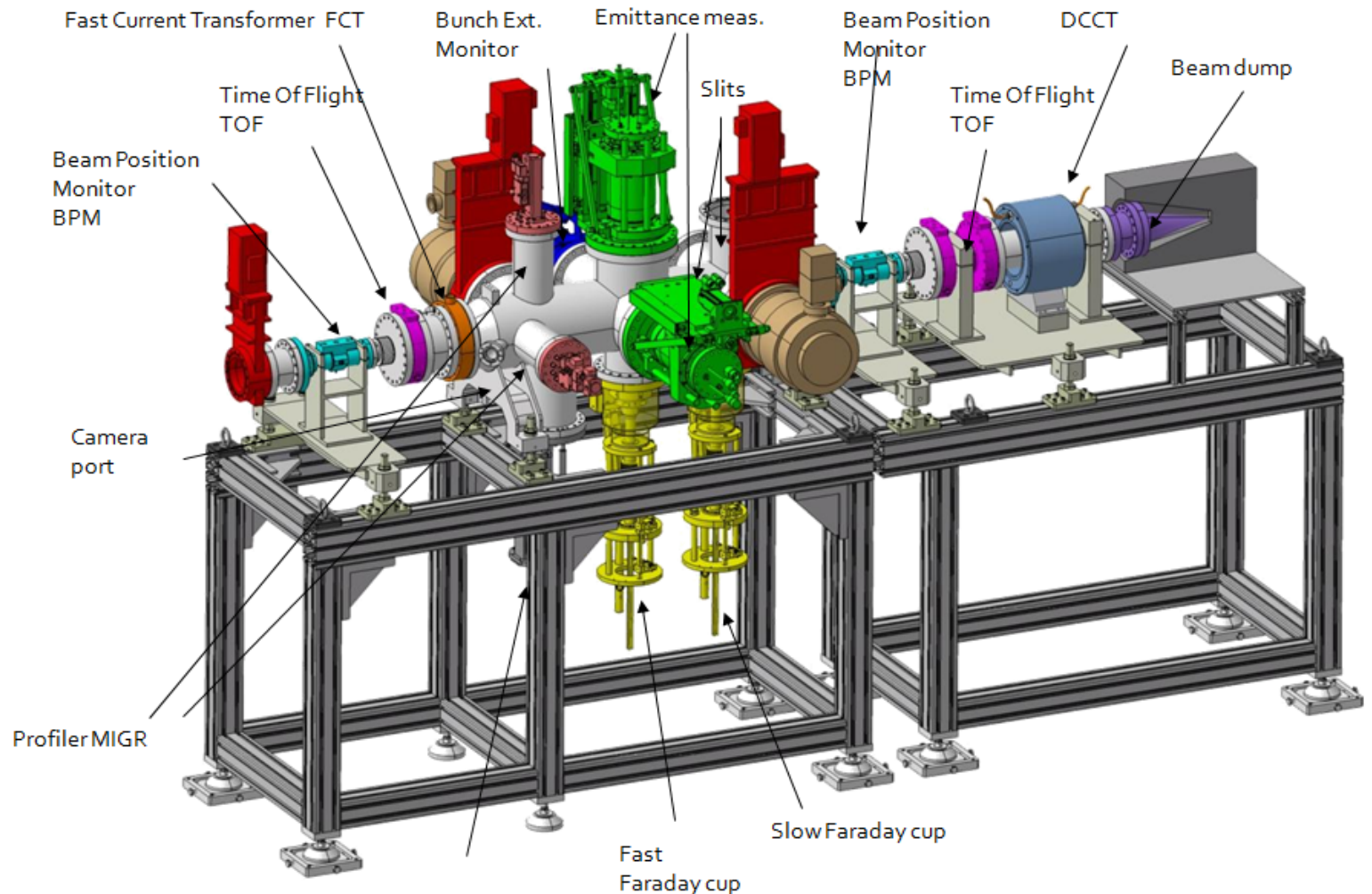


d state

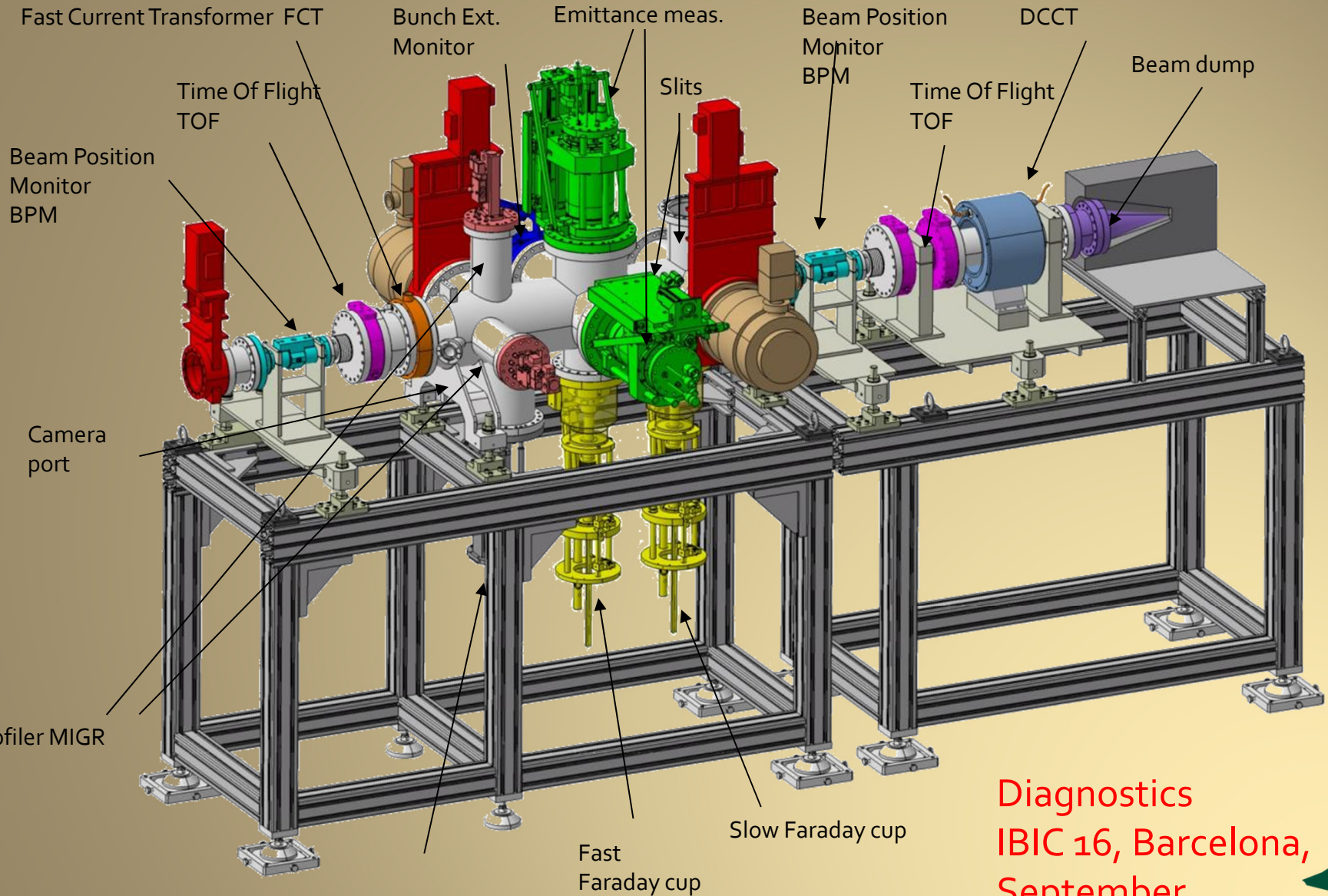
14kV

x-ups,

Test bench configuration



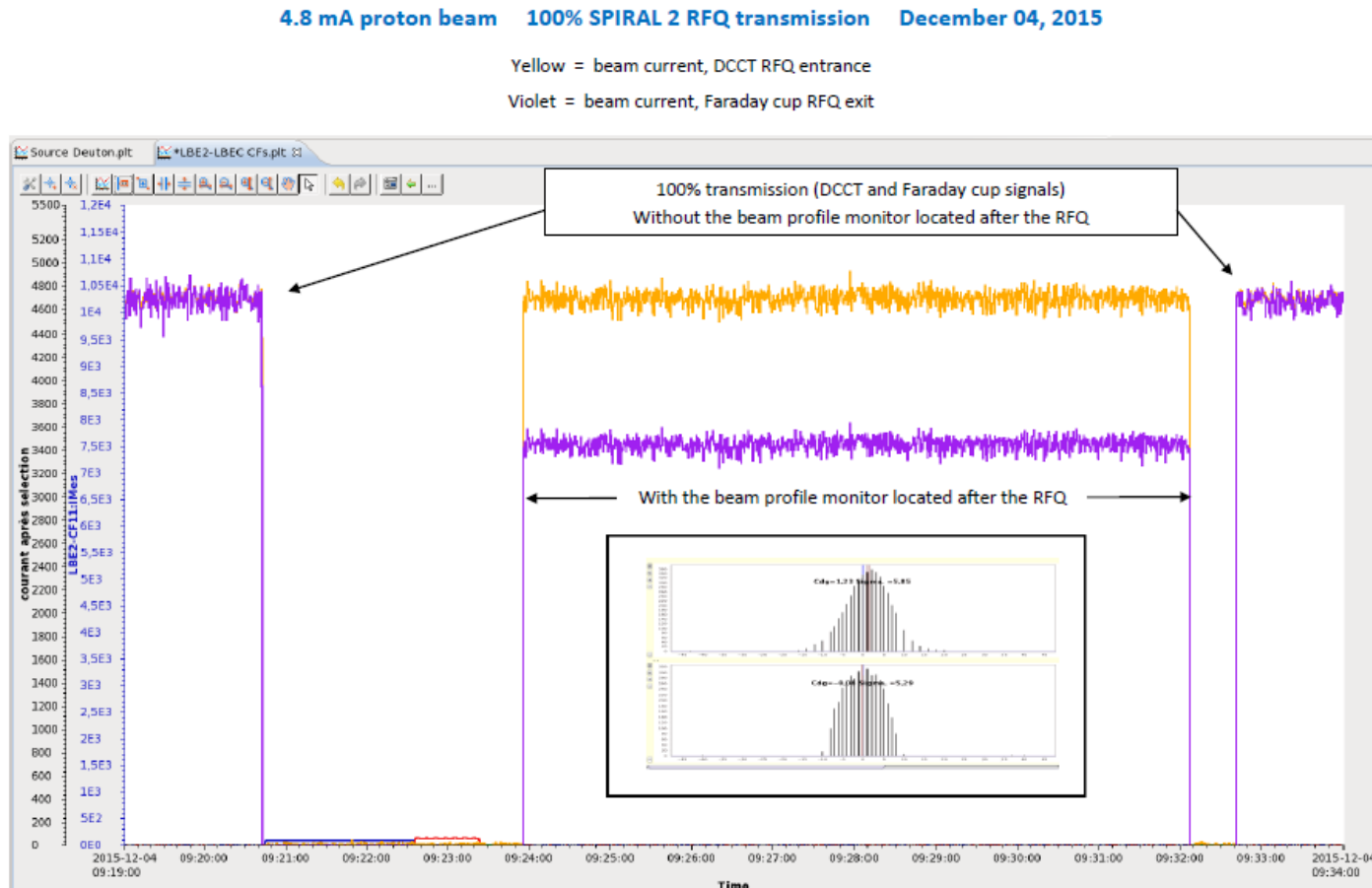
Injector test bench



Diagnostics
IBIC 16, Barcelona,
September

RFQ beam commissioning

- 2015, December 03, 9h26 : First RFQ beams (*Protons*)
- 4.8 mA with 100% transmission at the end of the same working day (SPIRAL 2 nominal beam current)



RFQ beam commissioning

First **CW proton beam** : December 18 : 2.3 mA

First **5 mA CW proton beam** : December 23, 2015

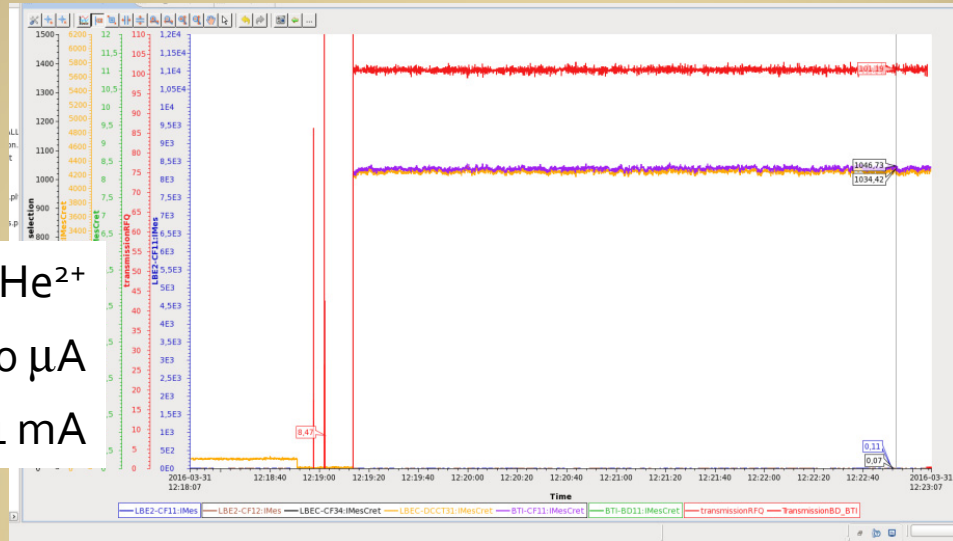
100 % transmission

Fine tuning using the RFQ vacuum

A/Q = 2 beam up to 1.5 mA $^4\text{He}^{2+}$

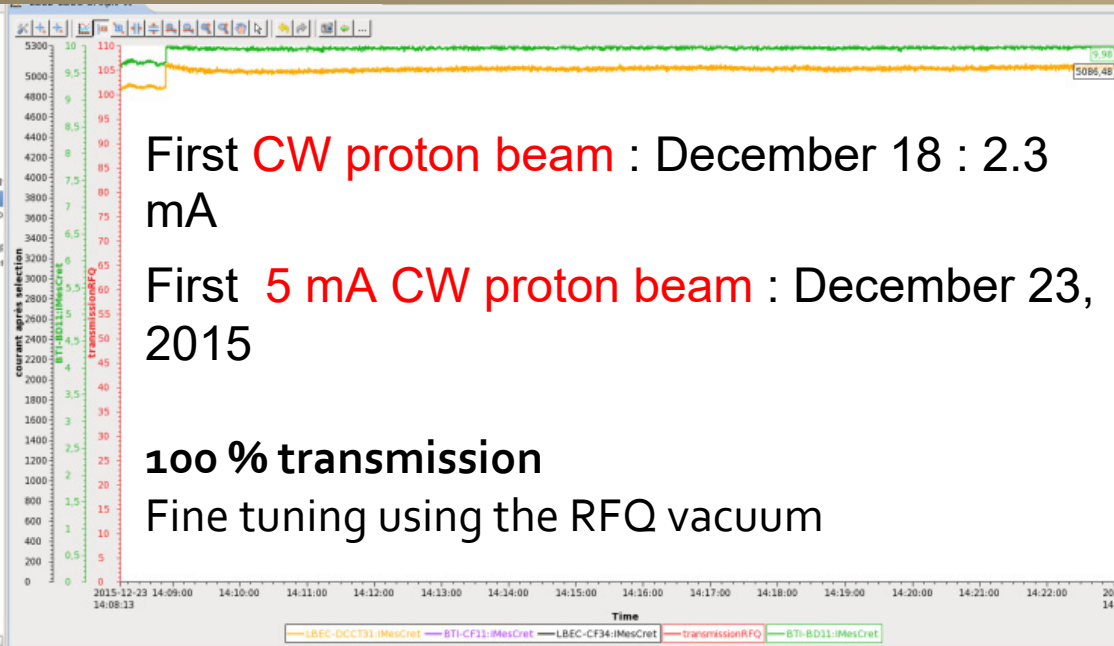
February 25, 2016 : First pulsed beam, 250 μA

First CW beam : June 04, 2016 ~ 1 mA



Proton & A/Q = 2 : RFQ always operated CW from the rf point of view
rf operation stable and reliable (with 3/4 amps, up to 80 kV)

RFO beam commissioning

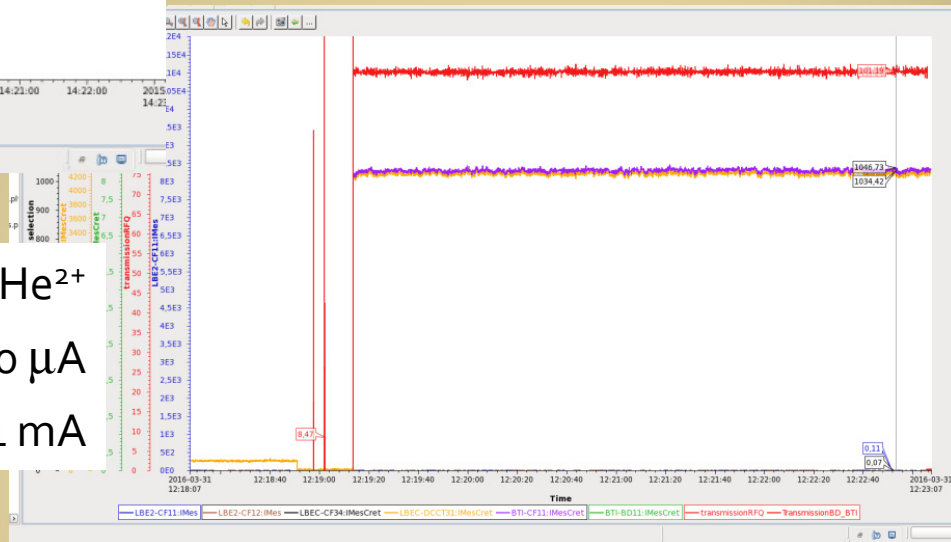


First **CW proton beam** : December 18 : 2.3 mA

First **5 mA CW proton beam** : December 23, 2015

100 % transmission

Fine tuning using the RFO vacuum



A/Q = 2 beam up to 1.5 mA $^4\text{He}^{2+}$

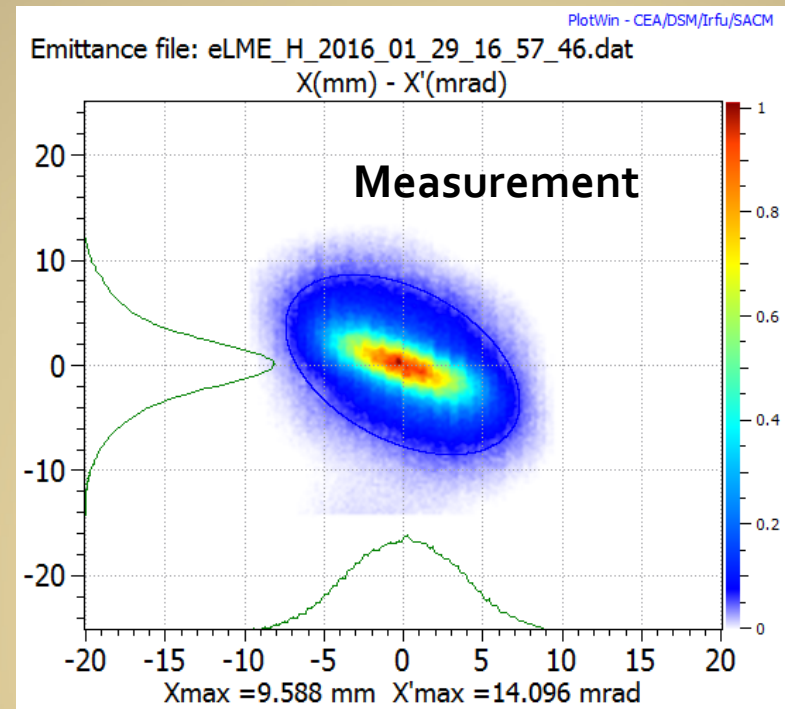
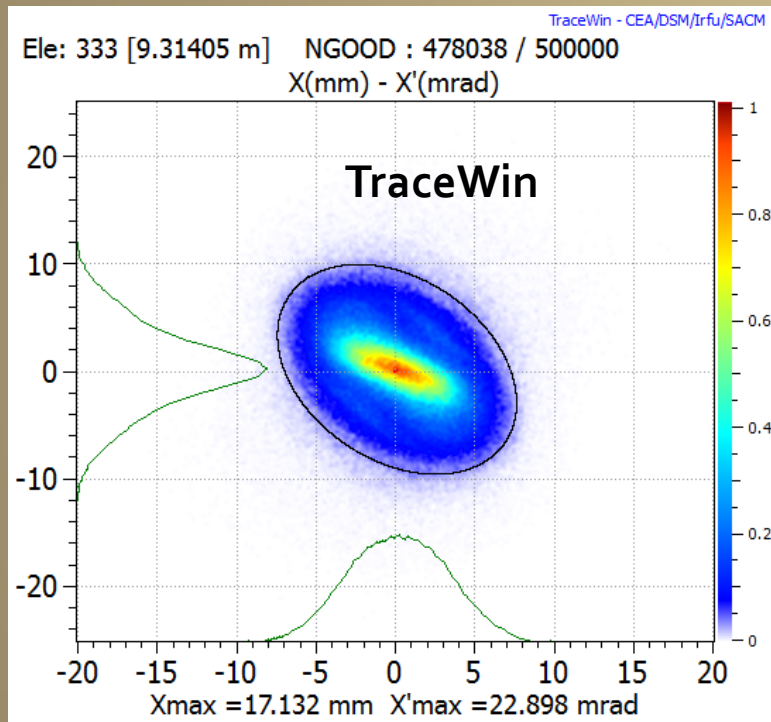
February 25, 2016 : First pulsed beam, 250 μA

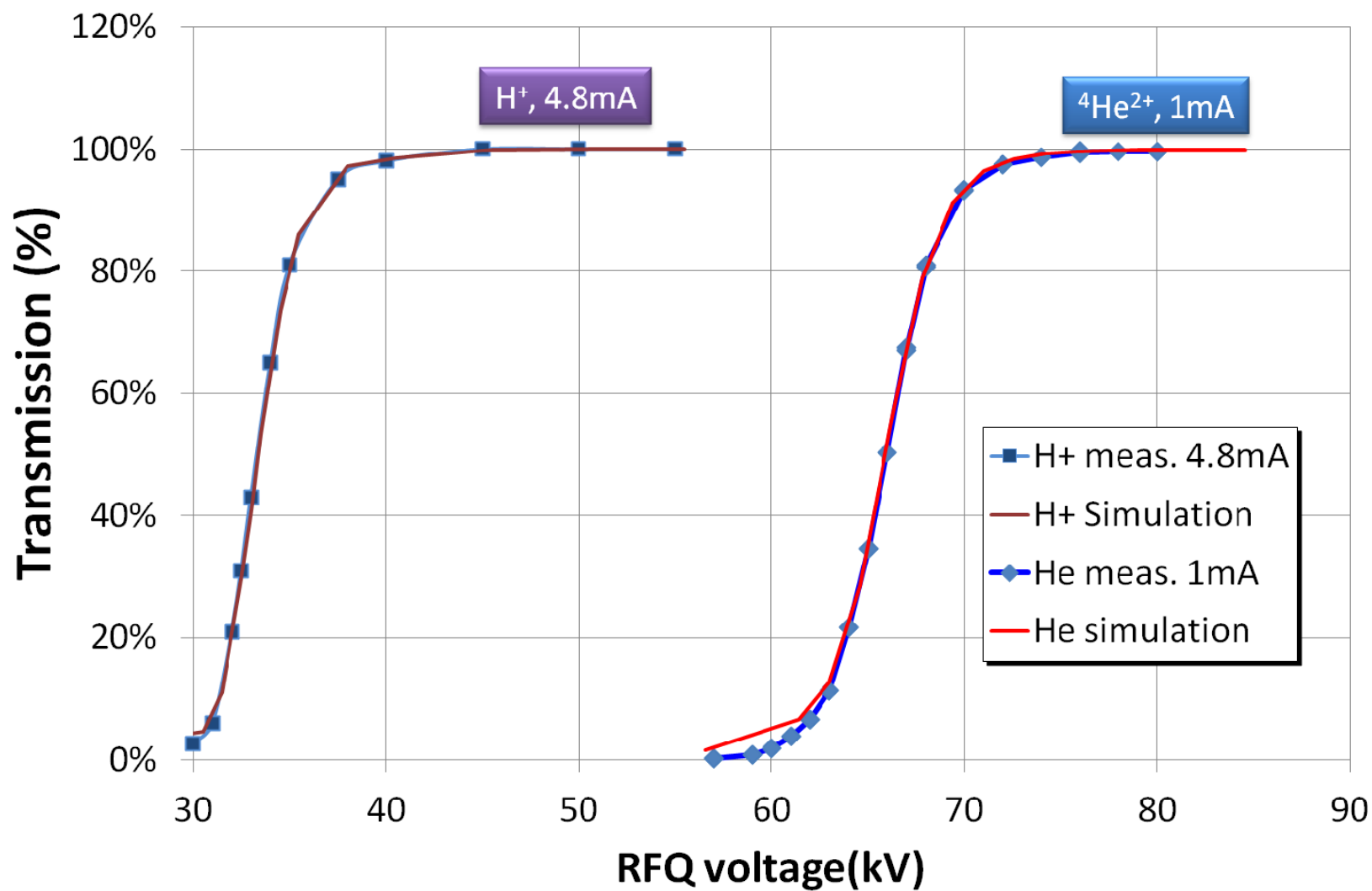
First CW beam : June 04, 2016 ~ 1 mA

Proton & A/Q = 2 : RFO always operated CW from the rf point of view
rf operation stable and reliable (with 3/4 amps, up to 80 kV)

RFO beam commissioning

Emittance measurement and comparison with the code





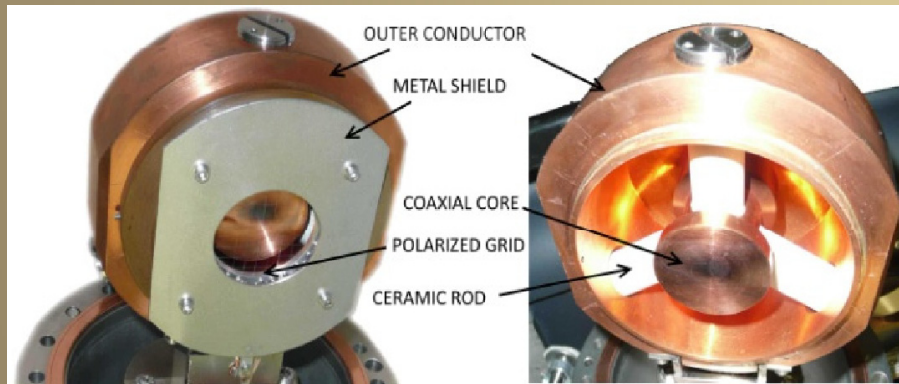
Longitudinal measurements

● TOF measurements (IBIC – W. LeCoz – WEPF32)

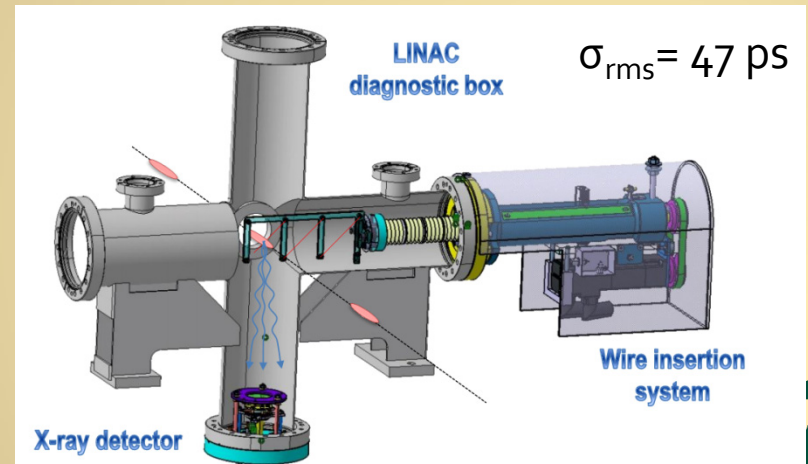
Energy (keV/nucleus)	Toutatis simulations	TOF measurement
Proton	730	729.3
Helium	727.2	728.1



● Fast Faraday Cup (IBIC – C. Jamet - WEPG42) and Beam Extension Measurement (IBIC – R. Revenko - TUPG59)

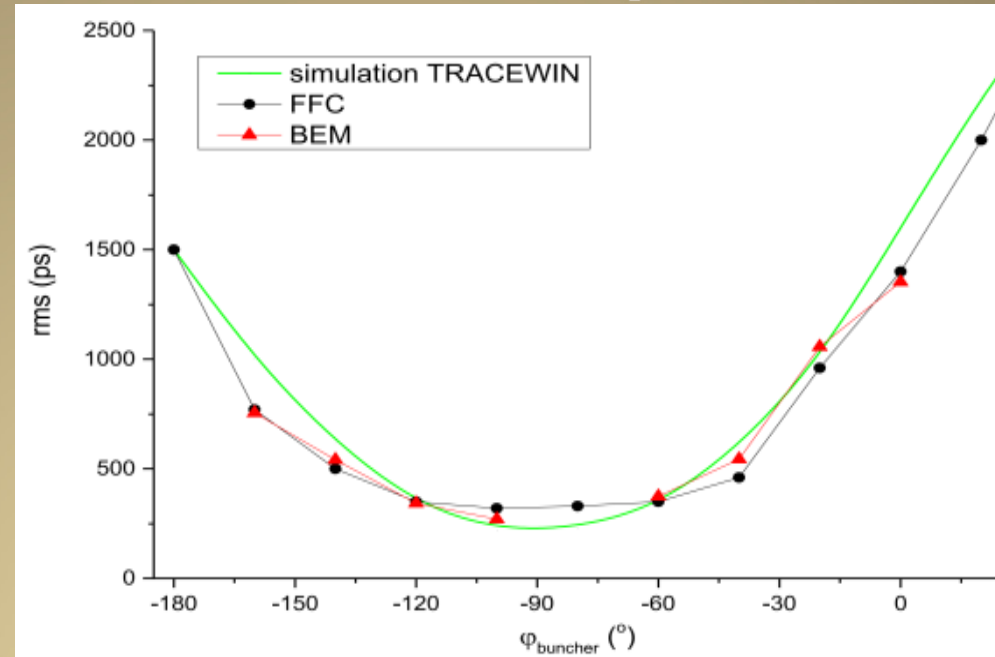


400W max, and $\sigma_{rms} = 320-330$ ps

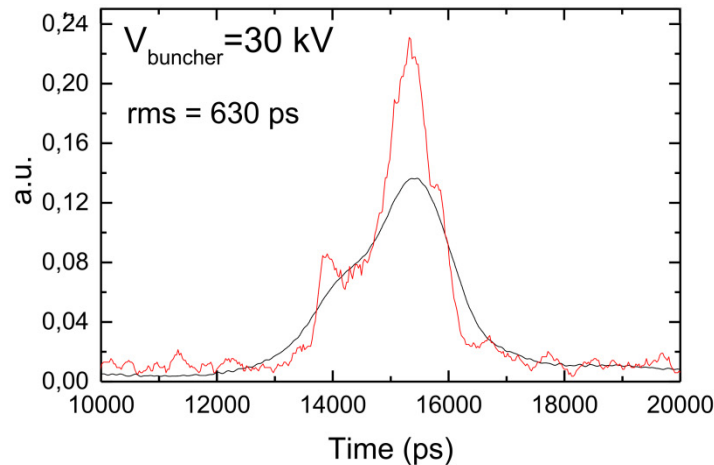
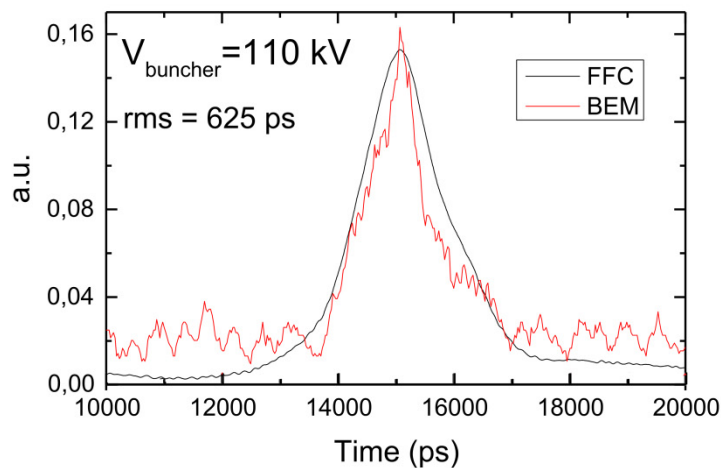


Bunch length measurement comparison

- good agreement between the two measurements and the TraceWin simulations
- Buncher variation in phase or voltage

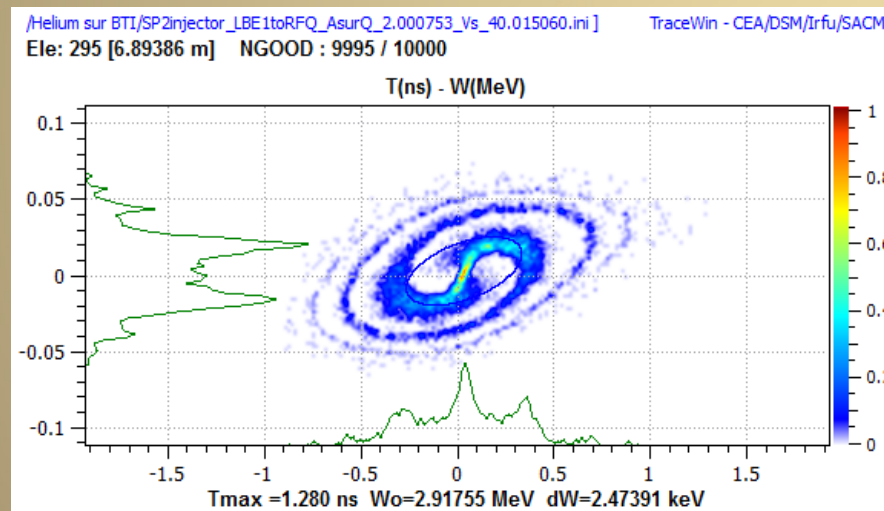
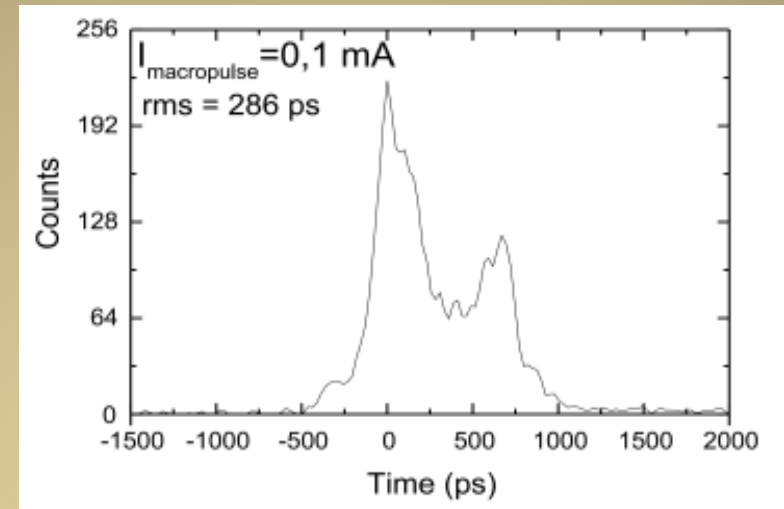
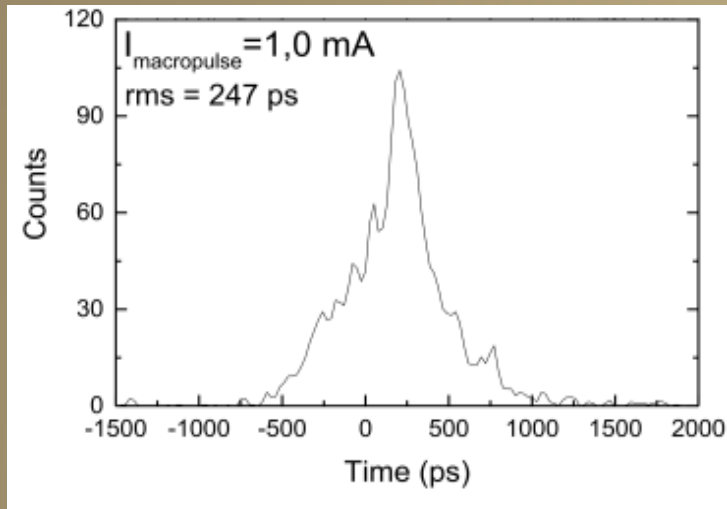


Comparison FFC with BEM, 0.6mA, 1%dC



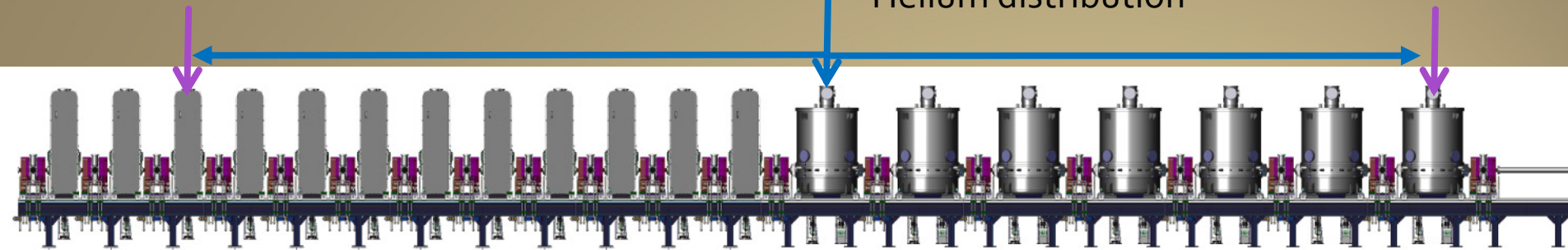
Longitudinal results

Very different shapes for 1 mA and $< 150\mu\text{A}$ beams



Linac first cool down

Helium distribution

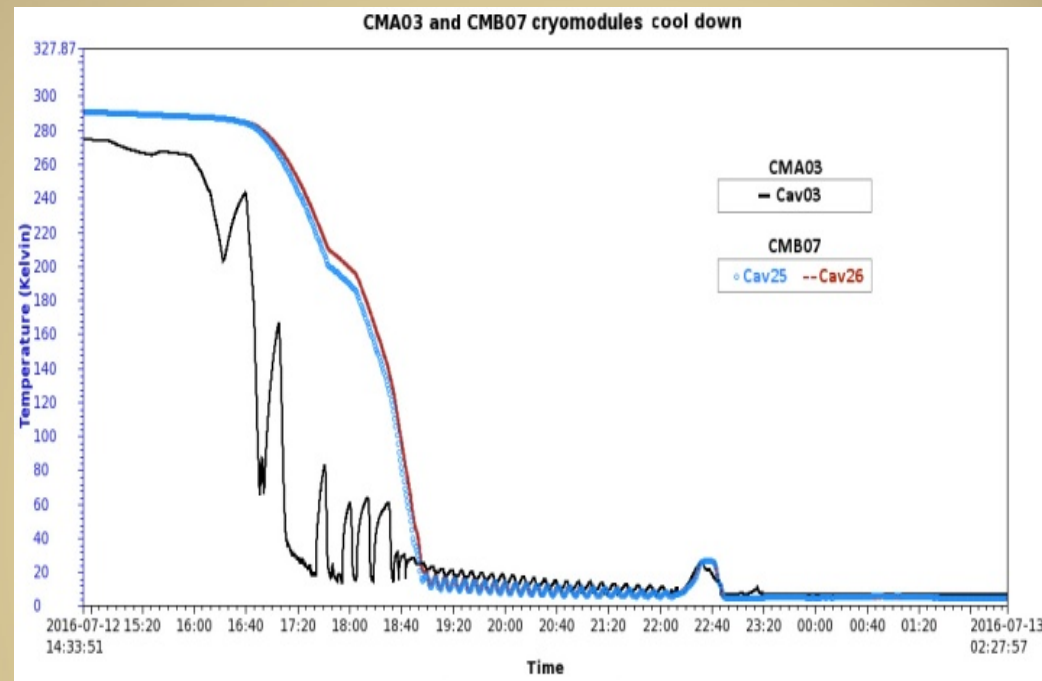


● First partial cool down on July 13, 2016

■ test a major part of the cryogenic installation

■ PLCs and C/C

● Both cryomodules were regulated at 4K after about 20 hours of cool-down



Partner labs for GANIL/SPIRAL2 phase 1

- CEA-Irfu (SACM, SIS, LENAC) : P/D source, RFQ, LEBT, A-type cryomodule, C/C, LLRF, Activation simulation
- CEA/DAM/ DP2I : safety
- CNRS-IN2P3 (IPNO, IPNL, IPHC, LPSC, LAL, LPC) : H1 source, B-type cryomodules, power couplers, diagnostics, final main beam bump, timing
- BARC : BPM electronic
- INFN : Slow Chopper, bunch selector (fast chopper), test stand
- IFJPAN: Valves boxes installation
- SOREQ : diagnostic beam test stand
- IFIN-HH : Beam loss monitor
- INRNE-BAS : linac supports
- Argonne : design of A/Q=1/6 injector
- CIEMAT : beam dump



Summary

- Good advances of the SPIRAL 2 project !!!
 - Both light and heavy ion sources working very well
 - RFQ working well with protons and $A/Q = 2$ ions (He)
 - ▶ RFQ with about 100% transmission
 - ▶ Very good comparisons with TraceWin simulations
- RFQ preliminary conditioned up to maximum voltage (114 kV, CW) ➤ $A/Q = 3$ ions soon
- RFQ improvements
 - rf amplifiers (reliability) LLRF (variable freq. loop) Tuning system (faster and stable)
- Superconducting linac : New cooling down in November
- RF in cryomodule : beginning of next year
- First linac beam : first semester of 2017





Thank you for
your attention

Thank all the CERN and partner labs staff

Thanks to P. Bertrand, M. Di Giacomo, H. Franberg, O. Kamalou, JM. Lagniel, G. Normand,
A. Savalle, F. Varenne, D. Uriot, JL. Biarrotte